Robust Certificates in Learning Theory: Combinatorial and Algorithmic Aspects

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Abstract

A robust certificate, in the framework of Blum, Hanneke, Pabbaraju, and Saless [BHPS25], is a small subset of training data that serves as a short, verifier-checkable proof of a prediction. We study two structural questions about such certificates, both of which lead to transversal– or hitting-set–type questions on natural hypergraphs derived from the hypothesis class. In [BHPS25], the authors also generalise this notion to the robust setting, where an adversary may maliciously corrupt up to b training samples.

Combinatorially, we give an upper bound on the length of b-robust certificates, equivalently on the *b-robust hollow star number* s_b, in terms of the unrobust hollow star number s_0 and the corruption budget b. Via a reduction to the tolerance framework for Helly-type theorems of Montejano and Oliveros [MO11], combined with Tuza's extremal estimate [Tuz89], we obtain a bound $s_b(\mathcal{H}) \leq \eta(s_0(\mathcal{H}), b{+}1)$, which specialises to $O(b^{\,s_0-1})$ for fixed s_0 and to $O(s_0^{b+1})$ for fixed b;

Algorithmically, we show that the problem of extracting a minimum-size certificate from a given training sample is exactly equivalent, in both directions, to finding a minimum transversal (equivalently, a minimum hitting set) in a hypergraph whose vertices are the sample occurrences in S, induced by the hypothesis class $\mathcal{H}$ and the training set $S \subseteq \mathcal{X} \times \mathcal{Y}$. We show that even under the assumption that the training set was obtained via the iterative sampling procedure proposed by [BHPS25], an arbitrary finite hypergraph can be induced, up to an i.i.d. blow-up and one forced singleton edge, with arbitrarily high probability. We further study an oracle-variant of the *certificate extraction* problem, and show that VC dimension d is sufficient for guaranteeing an $O(d \ln |S|)$-approximation algorithm using polynomially many oracle calls.

The two contributions are tied by a common hypergraph picture: such a pair $(\mathcal{H}, S)$ admits different hypergraph encodings, each natural to the problem at hand (the *certificate extraction* problem or the robust upper-bound question).

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1 Introduction

A central question in trustworthy machine learning is not only *whether* a model classifies a given input correctly, but *why*. Blum, Hanneke, Pabbaraju, and Saless [BHPS25] formalise one operational answer: a *certificate* for a prediction at a test point $x^\star$ is a small subset $C \subseteq S$ of the training data such that every classifier in the hypothesis class $\mathcal{H}$ consistent with C—or, in the robust version, consistent up to b label corruptions—predicts the same label on $x^\star$. The certificate is a short, verifier-checkable proof of the prediction, grounded in training data rather than in post-hoc heuristics. The classical example is Carathéodory's theorem: if $\mathcal{H}$ is the family of linear classifiers in $\mathbb{R}^d$ and $x^\star$ lies in the convex hull of $d+1$ positive training points, those $d+1$ points serve as a certificate for the positive prediction.

Hollow star number, dual Helly number, and b-robust variant. As shown in [BHPS25], the worst-case length of a minimal-inclusive certificate in this framework is characterized, up to the convention of whether the distinguished query point is counted, by the *hollow star number*. A labeled sequence S is a *hollow star set* for $\mathcal{H}$ if S is not realizable by $\mathcal{H}$ but every proper subsequence of S is; the hollow star number $s_0(\mathcal{H})$ is the supremum of $|S|$ over all such sequences. It was first introduced in Bousquet, Hanneke, Moran, and Zhivotovskiy [BHMZ20], and it is tightly related to their definition of *dual Helly number* of $\mathcal{H}$. If one restricts the *dual Helly number* to range only over *finite* collections, this results precisely in the *hollow star number* (made precise in Appendix B). The *b-robust* variant $s_b(\mathcal{H})$ extends the definition to a corruption budget of b label errors [BHPS25]. The question of how s_b grows in s_0 and b was left open in [BHPS25]:

Research Question 1. How does $s_b(\mathcal{H})$ grow with $s_0(\mathcal{H})$ and the corruption budget b?

The combinatorial bound (Section 3). Section 3 answers Research question 1 with a bound on $s_b(\mathcal{H})$ that is polynomial whenever *one* of $s_0(\mathcal{H})$ or b is held fixed (it is *not* polynomial if only the ratio s_0/b is fixed), obtained as a reduction to the tolerance framework for Helly-type theorems of Montejano and Oliveros [MO11] combined with the extremal estimate of Tuza [Tuz89]:

$$s_b(\mathcal{H}) \;\le\; \eta(s_0(\mathcal{H}),\, b{+}1) \;=\; \begin{cases} O(b^{\,s_0-1}) & \text{for fixed } s_0, \\ O(s_0^{\,b+1}) & \text{for fixed } b. \end{cases}$$

Both regimes follow from the same Tuza estimate $\eta(s_0, b{+}1) < \binom{s_0+b}{s_0-1} + \binom{s_0+b-1}{s_0-1}$;

Minimum certificate extraction (Section 4). A complementary question concerns the algorithmic side: given any certificate S for a prediction at $x^\star$, can we efficiently extract a *minimum-size* subset $C \subseteq S$ that is itself a certificate? We show (Section 4.3) that this problem is exactly equivalent, in both directions, to minimum transversal (equivalently, minimum hitting set) in a finite hypergraph on the sample occurrences of S: every triple $(\mathcal{H}, S, x^\star)$ induces such a hypergraph — the *mistake hypergraph* $\mathcal{M}$, whose edges are the mistake sets of bad-label hypotheses on S — and conversely every finite hypergraph on a finite ground set arises as the mistake hypergraph of some such triple. The extraction problem therefore inherits both the structural and the computational features of classical minimum transversal. We develop this only in the realizable ($b = 0$) setting; the b-robust analogue corresponds to a hitting set *with multiplicity* — each mistake set must be hit at least $b{+}1$ times — and is deferred to Section 6.

Two features of the learning context shape the algorithmic question. First, the learner may not have access to $\mathcal{H}$ explicitly; second, even the restriction $\mathcal{H}|_S$ may have size exponential in $|S|$, so any algorithm with running time $\Omega(|\mathcal{H}|_S|)$ is infeasible. We therefore study the problem in the *implicit hitting set* setting of Chandrasekaran, Karp, Moreno-Centeno, and Vempala [CKMCV11]: the hypergraph is accessed through a separation oracle (and the explicit hyperedges are hidden),

and complexity is measured in $|S|$ alone. We primarily analyse a (strong) weighted empirical-risk-minimization (ERM) separation oracle, which supports the LP-based algorithm of Section 4.5; weaker, more general oracle models are studied in [CKMCV11]. This raises:

Research Question 2. Can one give an algorithm that, in this oracle setting, extracts a minimum certificate efficiently (i.e. in time poly($|S|$)) — and to what approximation? Equivalently, are there triples $(\mathcal{H}, S, x^\star)$ for which no efficient algorithm can do so?

Our answer is partial. When the number $|\mathcal{M}|$ of distinct mistake sets is bounded, the LP oracle framework plus randomized rounding gives an $O(\log|\mathcal{M}|)$-approximation in polynomially many oracle calls (Section 4.5). Thus, if $|\mathcal{M}| \le |S|^{O(1)}$, the ratio becomes $O(\log|S|)$. This polynomial-row regime is reached, in particular, by classes of bounded VC dimension (yielding an $O(d\log|S|)$-approximation via Sauer–Shelah [Sau72, She72]); under bounded hollow star number we instead obtain an exact algorithm using $|S|^{O(s_0)}$ certificate-verification calls, hence polynomial time for fixed s_0. Whether a non-trivial approximation is achievable when $|\mathcal{M}|$ is super-polynomial is open.

Finally, we consider the *iterative sampling* framework of [BHPS25], in which a *certifier* — our term for an agent whose goal is to certify the label of a fixed test point $x^\star$ — acquires a certificate by drawing examples one at a time, stopping as soon as the sample certifies $x^\star$. We show that this stopping rule does not bypass the worst-case extraction barrier: even under it, an arbitrary finite hypergraph can be induced up to an i.i.d. blow-up and one forced singleton edge with arbitrarily high probability (Section 4.4), so iterative sampling alone does not constrain $\mathcal{M}$ enough to break the worst-case approximation barrier.

The hypergraph picture. In the *extraction* problem (Section 4) we give an explicit two-way reduction between our problem and minimum transversal in a hypergraph. The proof of the combinatorial bound (Section 3), through the tolerance Helly-type theorem of Montejano and Oliveros [MO11, Tuz89], also passes through a transversal-type question, but on a different extremal object (a b-critical, s_0-uniform hypergraph). We only note this surface resemblance: we do *not* establish any structural relationship between our mistake hypergraph and the hypergraph underlying the Montejano–Oliveros bound, and leave whether the two encodings can be unified as an open direction.

Contributions.

- An upper bound $s_b(\mathcal{H}) \le \eta(s_0(\mathcal{H}), b{+}1)$ on the b-robust hollow star number — equal to $O(b^{s_0-1})$ for fixed s_0 and $O(s_0^{b+1})$ for fixed b — via a reduction to the tolerance framework of Montejano and Oliveros [MO11] combined with Tuza's extremal estimate [Tuz89].
- An equivalence — in both directions — between minimum certificate extraction and minimum transversal in a finite hypergraph on S (Section 4.3), together with an $O(\log|\mathcal{H}|_S|)$-approximation algorithm in polynomially many weighted-ERM queries (Section 4.6), where $\mathcal{H}|_S$ is the restriction of $\mathcal{H}$ to S. In particular, this is $O(\log|S|)$ whenever $|\mathcal{H}|_S| \le |S|^{O(1)}$. Bounded VC dimension gives this polynomial-row regime, yielding an $O(d\log|S|)$-approximation; bounded hollow star number instead gives exact enumeration in polynomial time for fixed s_0.
- A side result on sampling-process dependence: even under the iterative sampling process of [BHPS25], an arbitrary finite hypergraph can be induced up to an i.i.d. blow-up and one forced singleton edge with arbitrarily high probability, so iterative sampling does not by itself constrain $\mathcal{M}$ enough to break the worst-case extraction barrier (Section 4.4).

Roadmap. Section 2 fixes notation and definitions. Section 3 proves the combinatorial bound as a clean reduction to the tolerance framework of Montejano and Oliveros. Section 4 introduces the mistake hypergraph, establishes the algorithmic equivalence with hypergraph transversal, and

develops its consequences (combinatorial assumptions, oracle implementation, sampling-process assumptions). Section 6 collects open directions. Appendix A gives an alternative, self-contained proof of the s_b bound that avoids the tolerance framework, together with a comparison of the two bounds.

2 Preliminaries

Setup and realizability. Let $\mathcal{X}$ be a domain and $\mathcal{Y} = \{0, 1\}$. A *hypothesis class* $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$ is a set of functions from $\mathcal{X}$ to $\mathcal{Y}$. A *labeled sequence* is a finite sequence $S = ((x_1, y_1), \ldots, (x_n, y_n)) \in (\mathcal{X} \times \mathcal{Y})^n$. We say that S is *realizable* by $\mathcal{H}$ if there exists $h \in \mathcal{H}$ with $h(x_i) = y_i$ for all $i \in [n]$. When we write $C \subseteq S$ for a sequence, we mean a subset of the index set $[n]$, or equivalently the corresponding subsequence. Thus repeated equal labeled examples are treated as distinct sample occurrences.

Version spaces, agreement and disagreement. For a labeled sequence S and a corruption budget $b \geq 0$, the *b-robust version space* of S is the set of hypotheses consistent with S up to b label errors,

$$V_b(S) \;=\; \Big\{h \in \mathcal{H} : \sum_{i=1}^{n} \mathbf{1}[h(x_i) \neq y_i] \;\leq\; b\Big\}.$$

The $b = 0$ case is the standard *version space* $V(S) := V_0(S) = \{h \in \mathcal{H} : h(x_i) = y_i \text{ for all } i\}$. We use the shorthand $V(x, y) := V(\{(x, y)\}) = \{h \in \mathcal{H} : h(x) = y\}$ for the version space of a singleton observation, so that $V(S) = \bigcap_i V(x_i, y_i)$ and realizability of S amounts to this intersection being nonempty.

For a version space $V \subseteq \mathcal{H}$, the *agreement region* of V and the *disagreement region* of V (the latter in the sense of Cohn, Atlas, and Ladner [CAL94]) are

$$\begin{aligned}\mathrm{Agr}(V) &= \{(x, y) \in \mathcal{X} \times \mathcal{Y} : h(x) = y \text{ for all } h \in V\},\\ \mathrm{DIS}(V) &= \{x \in \mathcal{X} : \exists\, h, h' \in V,\ h(x) \neq h'(x)\}.\end{aligned}$$

For a nonempty V the two notions are complementary: a point $x \in \mathcal{X}$ lies in $\mathrm{DIS}(V)$ iff *neither* $(x, 0)$ nor $(x, 1)$ lies in $\mathrm{Agr}(V)$; equivalently, $x \notin \mathrm{DIS}(V)$ iff V *decides* x, in which case exactly one of $(x, 0), (x, 1)$ belongs to $\mathrm{Agr}(V)$. We may abuse notation and write

$$\mathrm{Agr}_b(S) := \mathrm{Agr}(V_b(S)), \qquad \mathrm{DIS}_b(S) := \mathrm{DIS}(V_b(S)),$$

for the *b-robust* agreement and disagreement regions of a labeled sequence S, and $\mathrm{Agr}(S) := \mathrm{Agr}_0(S)$; the agreement region so defined matches [BHPS25, Definition 2.2].

Definition 2.1 (b-robust certificate, [BHPS25, Definition 2.3])**.** A subset $C \subseteq S$ is a *b-robust certificate* for a query $(x^\star, y^\star) \in \mathcal{X} \times \mathcal{Y}$ if C is b-robustly realizable ($V_b(C) \neq \varnothing$) and $(x^\star, y^\star) \in \mathrm{Agr}_b(C)$; that is, every hypothesis consistent with C up to b label errors predicts $y^\star$ at $x^\star$. The realizable case $b = 0$ gives the (standard) *certificate*.

The algorithmic study of minimum certificates is taken up in Section 4.

The (dual) Helly parameter and hollow stars. Recall that a finite family of sets has *Helly number* μ if, whenever every subfamily of size at most μ has a common point, the whole family does; classically (Helly's theorem) convex sets in $\mathbb{R}^d$ have Helly number $d + 1$. The parameter relevant here is the corresponding number for the version-space system $\{V(x, y) : (x, y) \in \mathcal{X} \times \mathcal{Y}\}$ of a hypothesis class.

Definition 2.2 (Dual Helly number, [BHMZ20, Definition 2])**.** The *dual Helly number* of a hypothesis class $\mathcal{H}$, denoted $k_w(\mathcal{H})$, is the smallest integer k such that, for every (possibly infinite) $S \subseteq \mathcal{X} \times \mathcal{Y}$ with $V(S) = \varnothing$, there exists a subset $W \subseteq S$ with $|W| \leq k$ and $V(W) = \varnothing$; if no such k exists, $k_w(\mathcal{H}) = \infty$. Equivalently, $k_w(\mathcal{H})$ is the Helly number of the set system $\{V(x, y) : (x, y) \in \mathcal{X} \times \mathcal{Y}\}$ on $\mathcal{H}$.

The *hollow star number* is the finite, obstruction-based refinement of this parameter: it measures the largest *minimal* unrealizable sequence.

Definition 2.3 (Hollow star)**.** Following [BHMZ20, Definition 3], a labeled sequence $S = ((x_1, y_1), \ldots, (x_m, y_m))$ is a *hollow star set* for $\mathcal{H}$ if

(i) S is not realizable by $\mathcal{H}$, and

(ii) for every $i \in [m]$, the subsequence $S \setminus \{i\}$ is realizable by $\mathcal{H}$.

The *hollow star number* $s_0(\mathcal{H})$ is the supremum of m for which such a sequence exists.

The hollow star number characterises the worst-case size of minimal-inclusive certificates in the realizable ($b = 0$) setting, up to the convention of whether the distinguished query point is counted [BHPS25, Theorem 3.7].

Remark 2.4 (Hollow star number as a finite refinement of the dual Helly number)*.* By [BHMZ20, Lemma 5], $s_0(\mathcal{H}) \leq k_w(\mathcal{H})$, with equality whenever $k_w(\mathcal{H}) < \infty$. The hollow star number can remain finite even when the dual Helly number is not: e.g. for the class of singletons over an infinite domain, $s_0 = 2$ while $k_w = \infty$. Restricting the quantifier in Definition 2.2 to *finite* collections recovers s_0 exactly (Proposition B.1); in this precise sense s_0 is the *finite refinement* of k_w.

Robust realizability and the b-robust hollow star. For a weighted labeled sequence $W = ((x_i, y_i, w_i))_{i=1}^m$ with positive integer weights, we say that W is *b-robustly realizable* by $\mathcal{H}$ if

$$\min_{h \in \mathcal{H}} \sum_{i=1}^{m} w_i \mathbf{1}[h(x_i) \neq y_i] \leq b.$$

Equivalently, after replacing each weighted point by w_i identical copies, some hypothesis realizes all but at most b copies.

Definition 2.5 (b-robust hollow star)**.** Following [BHPS25, Definition 3.4], a labeled sequence $T = ((x_1, y_1), \ldots, (x_m, y_m))$ is a *b-robust hollow star set* for $\mathcal{H}$ if there exist integer weights $w_1, \ldots, w_m$ and a distinguished index $i^\star \in [m]$ such that

(i) $w_i = 1$ for all $i \neq i^\star$, and $w_{i^\star} = b + 1$;

(ii) the weighted sequence is *not* b-robustly realizable: $\min_{h \in \mathcal{H}} \sum_{i=1}^{m} w_i \mathbf{1}[h(x_i) \neq y_i] > b$;

(iii) for every $j \in [m]$, removing (x_j, y_j, w_j) yields a b-robustly realizable weighted sequence.

The *b-robust hollow star number* $s_b(\mathcal{H})$ is the supremum of m for which such a sequence exists.

Setting $b = 0$ in Definition 2.5 recovers Definition 2.3: the weight of the star point becomes 1, and the condition $\min_h \sum_i w_i \mathbf{1}[h(x_i) \neq y_i] > 0$ is simply unrealizability.

Remark 2.6 (Certificate reading of a b-robust hollow star)*.* A b-robust hollow star set T admits a certificate interpretation in the sense of Definition 2.1. Let $C = \{(x_i, y_i) : i \neq i^\star\}$ be its light points. Condition (ii) forces every $h \in V_b(C)$ to satisfy $h(x_{i^\star}) \neq y_{i^\star}$: otherwise its weighted error on all of T would be at most its error on the light points, hence at most b. Hence $(x_{i^\star}, 1 - y_{i^\star}) \in \mathrm{Agr}_b(C)$, while condition (iii) applied to the removal of $i^\star$ gives $V_b(C) \neq \varnothing$. Thus C is a b-robust certificate for the query $(x_{i^\star}, 1 - y_{i^\star})$, and the single-point removals of (iii) make it minimal-inclusive.

Convention. Section 3 treats the general b-robust setting. The remainder of the document (Section 4) works in the realizable setting $b = 0$; extending those results to $b > 0$ is discussed in Section 6.

3 An upper bound on s_b via tolerance Helly

In this section we prove that

$$s_b(\mathcal{H}) \;\le\; \eta(s_0(\mathcal{H}),\, b{+}1) \;=\; \begin{cases} O(b^{\,s_0-1}) & \text{for fixed } s_0, \\ O(s_0^{\,b+1}) & \text{for fixed } b, \end{cases}$$

where η is the Erdős–Gallai extremal function for critical uniform hypergraphs. The high-level idea is to recast b-robust realizability as b-tolerant realizability of a monotone property, after which the tolerance framework for Helly-type theorems of Montejano and Oliveros [MO11] applies directly.

Hypergraph framing. This section works with the family of version-space sets $\{V(x, y) : (x, y) \in \mathcal{X} \times \mathcal{Y}\}$ on $\mathcal{H}$ (Section 2). By Definition 2.3 and the realizability dictionary $\bigcap_i V(x_i, y_i) \neq \varnothing$, the hollow star number $s_0(\mathcal{H})$ is the finite Helly number of this set system (see Appendix B), and a b-robust hollow star set is a tolerance-b obstruction to realizability. The Montejano–Oliveros reduction (Section 3.1) shows that tolerance Helly-type questions reduce, in any monotone setting, to vertex-cover questions on an associated uniform hypergraph; combined with Tuza's extremal bound for critical hypergraphs, this yields the displayed polynomial bound.

3.1 The tolerance Helly framework

Definition 3.1 (Monotone property and Helly number, [MO11])**.** Let F be a finite family and P a property on subfamilies of F. We say P is *monotone* if $P(F) \Rightarrow P(G)$ for all $G \subseteq F$. The *Helly number* of P is the smallest μ such that: if $P(G)$ holds for all $G \subseteq F$ with $|G| \le \mu$, then $P(F)$ holds. Equivalently, every inclusion-minimal subfamily of F for which P fails has size at most μ.

Definition 3.2 (k-tolerant version, [MO11])**.** A family F has property P *with tolerance* k if there exists $G \subseteq F$ with $|G| = k$ such that $F \setminus G$ has property P. We use the Montejano–Oliveros existential convention for tolerance: one is allowed to delete some k elements, not every set of k elements. For monotone properties, this is equivalent to allowing at most k deletions whenever the family has enough elements, since extra deletions preserve P.

Theorem 3.3 (Tolerance Helly bound, [MO11, Tuz89])**.** *If P is a monotone property with Helly number μ, then the b-tolerant version of P has Helly number at most $\eta(\mu, b{+}1)$. Here $\eta(\mu, k)$ is the* Erdős–Gallai extremal function*: the maximum number of vertices in a k-critical μ-uniform hypergraph, that is, a μ-uniform hypergraph whose minimum transversal has size k and whose transversal number strictly decreases when any single edge is deleted. By Tuza [Tuz89],*

$$\eta(\mu, k) \;<\; \binom{\mu + k - 1}{\mu - 1} + \binom{\mu + k - 2}{\mu - 1}.$$

3.2 Realizability as a monotone property

Let $\mathcal{H}$ be a hypothesis class. Define the property P on finite labeled sequences by

$$P(S) \iff \text{“}S \text{ is realizable by } \mathcal{H}\text{”}.$$

This is monotone: if S is realizable then so is every subsequence. By the equivalent characterisation in Definition 3.1, the Helly number of P is the largest size of an inclusion-minimal sequence on which P fails. Now P fails on S exactly when S is unrealizable, and such an S is inclusion-minimal iff every proper subsequence is realizable — which is precisely the definition of a hollow star set (Definition 2.3). The two conditions are thus equivalent, so the inclusion-minimal unrealizable sequences are exactly the hollow star sets and the Helly number of P is $s_0(\mathcal{H})$.

3.3 The reduction and main theorem

The key idea is to convert the weighted structure of a b-robust hollow star set into an unweighted sequence in which b-robust realizability corresponds to b-tolerance. Given a b-robust hollow star set T of size m, we construct the *unweighted lift* $T^{\uparrow}$ by duplicating the star point $b{+}1$ times. Since any deletion of at most b elements from $T^{\uparrow}$ leaves at least one star copy intact, the weighted error constraint is faithfully encoded as element-count tolerance.

Note that $T^{\uparrow}$ itself need not be inclusion-minimal among non-b-tolerant families—it may contain proper subsets that are also not b-tolerantly realizable (for instance with fewer star copies). Our strategy is therefore to show that $T^{\uparrow}$ *contains* an inclusion-minimal non-b-tolerant subfamily $F^{\star}$, and that any such $F^{\star}$ must include all $m-1$ light points. The Tolerance Helly theorem then bounds $|F^{\star}|$ and hence m.

Definition 3.4 (Unweighting via duplication)**.** Given a weighted sequence $T = \{(x_i, y_i, w_i)\}_{i=1}^{m}$, define the *unweighted multisequence* $T^{\uparrow}$ by replacing each (x_i, y_i, w_i) with w_i identical copies of (x_i, y_i).

By construction, T is b-robustly realizable if and only if $T^{\uparrow}$ is b-tolerantly realizable: the weighted error of any h on T equals the number of elements of $T^{\uparrow}$ that h misclassifies, and the monotonicity observation in Definition 3.1 lets us pass between "at most b" and the exact-b convention used in the tolerance definition.

Lemma 3.5. *Let T be a b-robust hollow star set for $\mathcal{H}$ of size m, with star point of weight $b{+}1$ and all other points of weight 1. Then $T^{\uparrow}$ is not b-tolerantly realizable; moreover, every subfamily of $T^{\uparrow}$ that is not b-tolerantly realizable must contain all $m-1$ light points.*

Proof. $T^{\uparrow}$ is not b-tolerantly realizable. Deleting at most b elements from $T^{\uparrow}$ leaves at least one copy of the star point. Any hypothesis realizing the remainder must classify the star point correctly, hence makes at most b additional mistakes on the light points—exactly a b-robust realization of T, contradicting the definition of a b-robust hollow star set.

Every non-b-tolerant subfamily contains all light points. Suppose a subfamily $F \subseteq T^{\uparrow}$ is missing some light point (x_i, y_i). Then $F \subseteq (T \setminus \{i\})^{\uparrow}$. Since $T \setminus \{i\}$ is b-robustly realizable, $(T \setminus \{i\})^{\uparrow}$ is b-tolerantly realizable, and by monotonicity of b-tolerance so is F. □

Theorem 3.6. $s_b(\mathcal{H}) \;\le\; \eta\big(s_0(\mathcal{H}),\, b{+}1\big)$.

Proof. Let T be a b-robust hollow star set of size m. By Lemma 3.5, $T^{\uparrow}$ is not b-tolerantly realizable for the monotone property P, whose Helly number is $s_0(\mathcal{H})$. By Theorem 3.3, $T^{\uparrow}$ contains an inclusion-minimal non-b-tolerant subfamily $F^{\star}$ with $|F^{\star}| \le \eta(s_0, b{+}1)$. By the second part of Lemma 3.5, $F^{\star}$ must contain all $m-1$ light points. It must also contain at least one copy of the star point: otherwise it would be contained in the lift of $T \setminus \{i^{\star}\}$, which is b-tolerantly realizable by condition (iii) of Definition 2.5. Hence $m \le |F^{\star}| \le \eta(s_0, b{+}1)$. □

Corollary 3.7 (Explicit bound and special regimes)**.** *For every hypothesis class $\mathcal{H}$ and every integer $b \ge 0$,*

$$s_b(\mathcal{H}) \;<\; \binom{s_0 + b}{s_0 - 1} + \binom{s_0 + b - 1}{s_0 - 1}.$$

In particular, $s_b(\mathcal{H}) = O(b^{\,s_0 - 1})$ when s_0 is held fixed, and $s_b(\mathcal{H}) = O(s_0^{\,b+1})$ when b is held fixed.

Proof. Combine Theorem 3.6 with Tuza's estimate for η (Theorem 3.3). The binomial $\binom{s_0+b}{s_0-1} = \binom{s_0+b}{b+1}$ is a polynomial of degree $s_0 - 1$ in b and of degree $b + 1$ in s_0, which gives the two regimes. □

We have just bounded, in the worst case over all hypothesis classes with a given s_0, the minimum size of an unavoidable certificate. An alternative, self-contained proof that does not invoke the Tolerance Helly theorem — at the cost of a looser bound — is given in Appendix A, where the two bounds are also compared (§A.3). Section 4 turns to the complementary algorithmic question: given a concrete sample S and test point $x^\star$, how do we efficiently *find* a small certificate inside S?

4 Certification extraction is hypergraph transversal

Fixing a sample S and a test point $x^\star$, we ask: given that the sample certifies $x^\star$, how do we efficiently extract a *small* subset of S that is also a certificate? We show that minimum certificate extraction is exactly minimum transversal (equivalently, minimum hitting set) in a hypergraph on the sample occurrences of S whose edges are the mistake sets of bad-label hypotheses. The equivalence holds in both directions, places the problem inside a classical optimisation template, and (Sections 4.5–4.6) yields concrete approximation guarantees and an LP-based implementation via a weighted-ERM oracle.

Throughout this section we work in the realizable setting $b = 0$. Extending the results to $b > 0$ is discussed in Section 6.

4.1 Problem setup

We work in the binary realizable setting. Let $\mathcal{H} \subseteq \{0,1\}^{\mathcal{X}}$ be a hypothesis class, and let

$$S = \big((x_1, y_1), \ldots, (x_n, y_n)\big) \in (\mathcal{X} \times \{0,1\})^n$$

be a realizable labeled sample. Fix a test point $x^\star \in \mathcal{X}$ and a label $y^\star \in \{0,1\}$, and specialise Definition 2.1 to $b = 0$: a subset $C \subseteq S$ is a *certificate* for $(x^\star, y^\star)$ if $V(C) \neq \varnothing$ and $(x^\star, y^\star) \in \mathrm{Agr}(C)$. We call C *minimal-inclusive* if no proper subset $C' \subsetneq C$ is also a certificate, and *minimum* if it has smallest cardinality among all certificates. Our optimization problem is

$$\mathrm{OPT} \;=\; \min\{|C| : C \subseteq S,\ C \text{ is a certificate for } (x^\star, y^\star)\},$$

and we write OPT for its optimal value, i.e. the minimum certificate size.

Throughout we assume S is itself a certificate for $(x^\star, y^\star)$, so that $S \cup \{(x^\star, y^\star)\}$ is realizable. Since realizability is inclusion-closed, every subset of $S \cup \{(x^\star, y^\star)\}$ is then realizable; in particular $C \cup \{(x^\star, y^\star)\}$ is realizable for every $C \subseteq S$, so the clause $V(C) \neq \varnothing$ holds automatically and being a certificate reduces to $C \cup \{(x^\star, 1 - y^\star)\}$ being *un*realizable — i.e. the wrong label is ruled out. It is therefore convenient to restrict attention to the *bad-label version space*

$$V^- \;=\; \{h \in \mathcal{H} : h(x^\star) = 1 - y^\star\},$$

since C is a certificate exactly when it excludes every $h \in V^-$. If $V^- = \varnothing$, then the empty set is already a certificate and the extraction problem is trivial. We therefore assume $V^- \neq \varnothing$ throughout the nontrivial algorithmic discussion below; under this assumption every certificate has size at least one.

4.2 Minimal-inclusive certificates are easy; minimum certificates are not

A minimal-inclusive certificate is easy to extract using only a realizability oracle.

Proposition 4.1. *Suppose we have oracle access that decides realizability with respect to $\mathcal{H}$. Then a minimal-inclusive certificate can be found in at most n oracle calls.*

Proof. Start with $C \leftarrow S$. Iterate over the points $z \in C$ in arbitrary order; for each, query whether $(C \setminus \{z\}) \cup \{(x^\star, 1 - y^\star)\}$ is realizable. If it is still not realizable, delete z; otherwise keep it. The output remains a certificate throughout (monotonicity of realizability), and at termination no remaining point can be deleted without destroying the certificate property. □

Remark 4.2. The worst-case size of a minimal-inclusive certificate coincides with the hollow star number s_0 up to an additive constant of 1 [BHPS25]; in particular, every minimal-inclusive certificate has size at most s_0. In the nontrivial case $V^- \neq \varnothing$, any certificate has size at least 1, the procedure of Proposition 4.1 is therefore an s_0-approximation algorithm for the minimum certificate problem. In the b-robust setting the same argument applies with s_b in place of s_0, giving an s_b-approximation.

The minimum certificate problem is significantly harder, as we now show.

4.3 Equivalence with hypergraph transversal

For $h \in V^-$, define the *mistake set* of h on S by

$$M(h, S) \;=\; \{z \in S : h(z_x) \neq z_y\},$$

where $z = (z_x, z_y)$. Since $S \cup \{(x^\star, 1 - y^\star)\}$ is unrealizable, every $h \in V^-$ satisfies $M(h, S) \neq \varnothing$. For convenience, we write $M(h)$ without S when it is clear from context.

Proposition 4.3 (Hitting-set reformulation)**.** *A subset $C \subseteq S$ is a certificate iff $C \cap M(h, S) \neq \varnothing$ for every $h \in V^-$.*

Proof. Since S is a certificate, $C \cup \{(x^\star, y^\star)\}$ is realizable for every $C \subseteq S$ (realizability is inclusion-closed), so C is a certificate iff $C \cup \{(x^\star, 1 - y^\star)\}$ is unrealizable, i.e. iff no $h \in V^-$ is consistent with C. If C is a certificate, every $h \in V^-$ mistakes at least one point in C; otherwise h would realize $C \cup \{(x^\star, 1 - y^\star)\}$. Conversely, if $C \cap M(h, S) \neq \varnothing$ for all $h \in V^-$, then no bad-label hypothesis is consistent with C, so $C \cup \{(x^\star, 1 - y^\star)\}$ is unrealizable. □

The minimum certificate problem is therefore the minimum hitting set problem for the family $\{M(h, S) : h \in V^-\}$, which we package as a hypergraph on the ground set S.

Definition 4.4 (Hypergraph transversal)**.** A *hypergraph* on a ground set V is a family $\mathcal{F} \subseteq 2^V$ of nonempty *edges*. A *transversal* (or *hitting set*, or *vertex cover*) of $\mathcal{F}$ is a set $T \subseteq V$ with $T \cap E \neq \varnothing$ for every $E \in \mathcal{F}$; the *minimum transversal problem* asks for such a T of smallest cardinality. A hypergraph is *simple* (a *clutter*) if no edge contains another.

Definition 4.5 (Mistake hypergraph)**.** The *mistake hypergraph* of $(\mathcal{H}, S, x^\star, y^\star)$ is

$$\mathcal{M} \;=\; \{M(h, S) : h \in V^-\} \;\subseteq\; 2^S.$$

Remark 4.6 (Only Inclusion-Minimal Edges Matter). Inclusion-larger edges are redundant for hitting: if $M(h) \subseteq M(h')$, then any set hitting $M(h)$ also hits $M(h')$. Restricting $\mathcal{M}$ to its inclusion-minimal edges — which form a simple hypergraph (clutter) — leaves the family of feasible transversals, and hence the optimum, unchanged.

The next example shows that although one might be tempted to keep only hypothesises which are minimizers on S (i.e. $|M(h, S)| = \min_{h \in H} |M(h', S)|$), this fails in general.

Example 4.7. Consider

	z_1	z_2	z_3	z_4
h_1	1	0	0	1
h_2	0	1	0	1
h_3	1	1	1	0

i.e. $M(h_1) = \{z_1, z_4\},\ M(h_2) = \{z_2, z_4\},\ M(h_3) = \{z_1, z_2, z_3\}.$

The minimum row size is 2, attained by h_1 and h_2; these are the wrong-label ERMs. Yet $M(h_3)$ is still inclusion-minimal—neither $M(h_1)$ nor $M(h_2)$ is contained in it—and h_3 cannot be dropped: $\{z_4\}$ hits both ERM rows but misses $M(h_3)$. In particular, $\mathcal{M}$ need not be a uniform hypergraph, even after pruning to its inclusion-minimal edges.

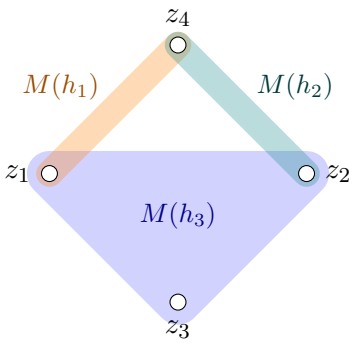


Figure 1: The mistake hypergraph of Example 4.7 on $S = \{z_1, z_2, z_3, z_4\}$. It is non-uniform even after pruning to inclusion-minimal edges: the 2-edges $M(h_1) = \{z_1, z_4\}$, $M(h_2) = \{z_2, z_4\}$ and the 3-edge $M(h_3) = \{z_1, z_2, z_3\}$ are pairwise incomparable. The vertex z_4 hits both 2-edges but misses $M(h_3)$, so no singleton is a transversal; a minimum transversal has size 2 (e.g. $\{z_3, z_4\}$).

Proposition 4.8. *The minimum certificate problem is exactly the minimum transversal problem in the mistake hypergraph $\mathcal{M}$.*

Proof. By Proposition 4.3, $C \subseteq S$ is a certificate iff it intersects every $M(h)$, $h \in V^-$, i.e. iff C is a transversal of $\mathcal{M}$. The two minimisation problems therefore coincide. □

We now prove the converse: every finite hypergraph arises this way.

Proposition 4.9. *Let $\mathcal{F} \subseteq 2^{[m]}$ be any finite hypergraph with no empty edge (the edges need not be inclusion-incomparable). Then there is a certificate-extraction instance $(\mathcal{H}, S, x^\star, y^\star)$ whose mistake hypergraph is exactly $\mathcal{F}$.*

Proof. Let $\mathcal{X} = \{x_1, \ldots, x_m, x^\star\}$, $S = ((x_1, 0), \ldots, (x_m, 0))$, and $y^\star = 0$, and set $h^\star \equiv 0$ on $\mathcal{X}$. For each edge $E \in \mathcal{F}$ define

$$h_E(x^\star) = 1, \qquad h_E(x_i) = \mathbf{1}[i \in E] \quad (i \in [m]),$$

and let $\mathcal{H} = \{h^\star\} \cup \{h_E : E \in \mathcal{F}\}$. Then $S \cup \{(x^\star, 0)\}$ is realized by $h^\star$, while $S \cup \{(x^\star, 1)\}$ is unrealizable: every hypothesis labelling $x^\star$ by 1 is some h_E, and since $\varnothing \notin \mathcal{F}$ each h_E makes at least one mistake on S. Hence $V^- = \{h_E : E \in \mathcal{F}\}$ and $M(h_E) = \{(x_i, 0) : i \in E\}$, so the mistake hypergraph is $\mathcal{M} = \{M(h) : h \in V^-\} = \mathcal{F}$ up to the relabelling $x_i \leftrightarrow i$. This holds verbatim whether or not $\mathcal{F}$ has nested edges; simplicity is never used. □

Corollary 4.10. *In the unrestricted static setting, minimum certificate extraction and minimum hypergraph transversal (equivalently, minimum hitting set) are equivalent optimization problems, in both directions.*

Remark 4.11. This is a structural equivalence; it does not by itself imply an oracle-efficient reduction in the learning model. Section 4.5 addresses the implementation question.

4.4 Iterative sampling assumption

The equivalence of Section 4.3 shows that, in the worst case over $(\mathcal{H}, S, x^\star)$, minimum certificate extraction is as hard as minimum hypergraph transversal. One might hope that the way the sample S is *acquired* restricts the induced hypergraph $\mathcal{M}$ enough to bypass this barrier. We show that the iterative sampling process of [BHPS25] does not: even there, essentially any hypergraph can be induced with arbitrarily high probability.

In this model a *certifier* draws labeled examples i.i.d. from a distribution $\mathcal{D}$ over $\mathcal{X}$ (labeled by a fixed target $h^\star \in \mathcal{H}$) and stops at the first time T at which the sample $S_T = ((x_1, y_1), \ldots, (x_T, y_T))$ certifies $x^\star$. The question is whether this stopping rule prevents arbitrary hypergraph structure from arising on S_T. The answer is essentially no: stopping forces a single *singleton* edge — corresponding to the last sample drawn — but after removing it the residual hypergraph can be an i.i.d. blow-up of any prescribed hypergraph.

Definition 4.12 (i.i.d. blow-up)**.** Let $\mathcal{C} \subseteq 2^{[m]}$ be a hypergraph. An *i.i.d. blow-up* of $\mathcal{C}$ is a hypergraph obtained by replacing each ground element $i \in [m]$ by a nonempty block G_i of sample occurrences (the draws equal to a fixed point x_i), the blocks being pairwise disjoint, and replacing each edge $E \in \mathcal{C}$ by $\bigcup_{i \in E} G_i$. Since the blocks are nonempty and disjoint, the map $E \mapsto \bigcup_{i \in E} G_i$ preserves (non-)containment, so the blow-up has the same transversal structure as $\mathcal{C}$.

Proposition 4.13. *Let $\mathcal{C} \subseteq 2^{[m]}$ be any finite hypergraph with no empty edge. For every $\delta > 0$ there exist a hypothesis class $\mathcal{H}$, a target $h^\star \in \mathcal{H}$, a test point $x^\star$, and a distribution $\mathcal{D}$ over $\mathcal{X}$ such that the following holds. Draw samples i.i.d. from $\mathcal{D}$, labeled by $h^\star$, and stop at the first time T at which S_T certifies $(x^\star, h^\star(x^\star))$. Then with probability at least $1 - \delta$ the induced mistake hypergraph on S_T is*

$$\mathcal{M} \;=\; \{\{z_T\}\} \cup \widetilde{\mathcal{C}},$$

where $\{z_T\}$ is the singleton edge of the last sample and $\widetilde{\mathcal{C}}$ is an i.i.d. blow-up of $\mathcal{C}$.

Proof. Let $\mathcal{X} = \{x_1, \ldots, x_m, x_\dagger, x^\star\}$, where $x_\dagger$ is a *rare trigger*. Set the target $h^\star \equiv 0$ on $\mathcal{X}$, so $y^\star = h^\star(x^\star) = 0$. For each edge $E \in \mathcal{C}$ define a bad-label hypothesis h_E by

$$h_E(x^\star) = 1, \qquad h_E(x_\dagger) = 0, \qquad h_E(x_i) = \mathbf{1}[i \in E] \;\; (i \in [m]),$$

and one further bad-label hypothesis $h_\dagger$ by

$$h_\dagger(x^\star) = 1, \qquad h_\dagger(x_i) = 0 \;\; (i \in [m]), \qquad h_\dagger(x_\dagger) = 1.$$

Let $\mathcal{H} = \{h^\star\} \cup \{h_E : E \in \mathcal{C}\} \cup \{h_\dagger\}$, and define $\mathcal{D}$ by $\mathcal{D}(x^\star) = 0$, $\mathcal{D}(x_\dagger) = q$ and $\mathcal{D}(x_i) = (1-q)/m$ for $i \in [m]$, with $q > 0$ chosen below.

Put

$$N = \left\lceil m \log\left(\frac{2m}{\delta}\right) \right\rceil \qquad \text{and choose} \qquad q \le \frac{\delta}{2N}.$$

First, the probability that a trigger appears among the first N samples is at most

$$\Pr[\exists t \le N : x_t = x^\dagger] \le \sum_{t=1}^{N} \Pr[x_t = x^\dagger] = Nq \le \frac{\delta}{2},$$

so with probability at least $1 - \delta/2$, no trigger appears during the first N draws.

Conditioned on this event, each of the first N samples is drawn independently from the conditional distribution of D on $\{x_1, \ldots, x_m\}$, which is uniform. Therefore, for any fixed $i \in [m]$,

$$\Pr[x_i \text{ does not appear among the first } N \text{ samples} \mid \text{no trigger in the first } N] = \left(1 - \frac{1}{m}\right)^N \le e^{-N/m}.$$

By the union bound over $i \in [m]$,

$$\Pr[\exists i \in [m] : x_i \text{ does not appear among the first } N \text{ samples} \mid \text{no trigger in the first } N] \le me^{-N/m} \le me^{-\log(2n$$

Thus, with probability at least $1-\delta/2$ conditioned on no trigger in the first N draws, all ordinary points appear among the first N samples. Combining the two estimates,

$$\Pr[\text{all ordinary points appear before the first trigger}] \ge 1 - \frac{\delta}{2} - \frac{\delta}{2} = 1 - \delta.$$

Call this event $\mathcal{E}$.

On $\mathcal{E}$, before the first trigger the hypothesis $h_\dagger$ is consistent with every observed sample (it labels each x_i by 0) and has $h_\dagger(x^\star) = 1$, so the wrong label at $x^\star$ remains realizable and the process cannot stop. At the first trigger, $h_\dagger$ is eliminated (since $h_\dagger(x_\dagger) = 1 \neq 0$); and since every $E \in \mathcal{C}$ is nonempty and all ordinary points have already appeared, each h_E has also made a mistake on some sampled point and is eliminated. Thus certification occurs exactly at the first trigger time T.

For $i \in [m]$ let $G_i = \{t \le T : x_t = x_i\}$; on $\mathcal{E}$ each G_i is nonempty, and since T is the first trigger time the trigger block is $\{T\}$. Computing mistake sets on S_T,

$$M(h_\dagger) = \{z_T\}, \qquad M(h_E) = \bigcup_{i \in E} G_i \quad (E \in \mathcal{C}).$$

Because the blocks G_i are nonempty and pairwise disjoint, the map $E \mapsto \bigcup_{i\in E} G_i$ preserves (non-)containment, so $\widetilde{\mathcal{C}} = \{\bigcup_{i\in E} G_i : E \in \mathcal{C}\}$ faithfully represents $\mathcal{C}$; together with the singleton $\{z_T\}$ (disjoint from all its edges), the induced mistake hypergraph is therefore $\mathcal{M} = \{\{z_T\}\} \cup \widetilde{\mathcal{C}}$. □

Corollary 4.14. *Iterative stop-on-certification sampling forces a single singleton edge, but this is essentially the only universal obstruction: after deleting it, the residual hypergraph can be an i.i.d. blow-up of an arbitrary prescribed hypergraph.*

Remark 4.15. The blow-up does not ease the transversal problem. Writing $\tau(\cdot)$ for the minimum transversal number, any transversal of $\widetilde{\mathcal{C}}$ projects to a transversal of $\mathcal{C}$, and any transversal of $\mathcal{C}$ lifts by picking one representative from each nonempty block G_i; hence $\tau(\widetilde{\mathcal{C}}) = \tau(\mathcal{C})$, and the stopped instance has

$$\tau\big(\{\{z_T\}\} \cup \widetilde{\mathcal{C}}\big) = 1 + \tau(\mathcal{C}).$$

So iterative sampling alone does not constrain $\mathcal{M}$ enough to break the worst-case hypergraph-transversal barrier of Section 4.3.

4.5 Oracle-based algorithm

GenAI Usage. *I used ChatGPT [Ope26] informally while learning and refining the ideas regarding the LP-oracle approach developed in this subsection.*

While our previous results mainly show a conceptual equivalence, they do not provide an applicable algorithm in the context of *learning theory*. Commonly, a learner does not have direct access to the hypothesis class $\mathcal{H}$ and may not be able to access the incidence matrix of the mistake hypergraph that the explicit-input algorithms of Appendix C assume. Furthermore, even if a learner does have access to such a matrix, since the number of edges may be exponential in $|S|$, any algorithm that runs in $\Omega(\text{poly}(|E|))$ time may be infeasible (merely verifying that a given subset is a hitting set may take $\Omega(|E|)$). This motivates oracle-based frameworks. The common framework is querying an oracle with a chosen subset $C \subseteq S$ and observing an answer, e.g. verification (true iff C is a hitting set). While different oracle assumptions exist, we primarily analyse a rather strong oracle, a weighted empirical-risk-minimization (ERM) oracle, which

allows us to use an LP-relaxation algorithm. We refer the reader to [CKMCV11] where weaker (more general) oracles are studied for specific problem instances (e.g. the minimum feedback vertex set problem on random graphs).

The plan of this subsection is to extract a small certificate without ever materialising the (possibly exponentially many) mistake-set constraints, in four steps.

(1) By Proposition 4.3, the minimum certificate problem is a hitting-set LP with only $|S|$ variables but one constraint per $h \in V^-$.
(2) A weighted bad-label ERM oracle acts as a separation oracle for this LP: its minimum-weight mistake set either exhibits a violated constraint or certifies feasibility. (In fact, a weighted ERM is stronger than a separation oracle).
(3) By the Grötschel–Lovász–Schrijver optimization-by-separation theorem (Theorem 4.17), the LP is then solved with roughly poly($|S|$) oracle calls, independently of the number of constraints — we use the ellipsoid method purely as a black box.
(4) Randomized rounding (Lemma C.2) turns the fractional optimum into a certificate, with approximation ratio $O(\log M)$ whenever M bounds the number of distinct mistake sets.

We now develop each step.

Assumptions. Throughout this subsection we assume each oracle query runs in time poly($|S|$), independently of the size or explicit representation of $\mathcal{H}$, and call an algorithm *efficient* if its total running time is poly($|S|$).

For any fixed constant $\rho \in (0, 1]$, a $\rho|S|$-approximation can be computed efficiently using only certificate verification, by brute-force enumeration of all subsets of size at most $\lceil 1/\rho \rceil - 1$ (Proposition C.3). The meaningful target is therefore not a constant-factor improvement over the trivial $|S|$-approximation, but a genuinely sublinear ratio such as $O(\log |S|)$.

The hitting-set LP. By Proposition 4.3, the minimum certificate problem is the integer program

$$\min \sum_{z \in S} x_z$$

subject to

$$\sum_{z \in M(h)} x_z \geq 1 \qquad \text{for every } h \in V^-,$$

and

$$x_z \in \{0, 1\} \qquad (z \in S).$$

Its LP relaxation is

$$\min \sum_{z \in S} x_z$$

subject to

$$\sum_{z \in M(h)} x_z \geq 1 \qquad \text{for every } h \in V^-,$$

and

$$0 \leq x_z \leq 1 \qquad (z \in S).$$

The upper bounds $x_z \leq 1$ are harmless: truncating any feasible solution at 1 cannot increase the objective and cannot violate any covering constraint. They are useful because they place the feasible region inside the explicit box $[0, 1]^S$, which is needed for the standard ellipsoid runtime bound. We now recall the standard optimization-by-separation framework used by the ellipsoid method.

Definition 4.16 (Separation oracle)**.** Let $P \subseteq \mathbb{R}^d$ be a convex set. A *separation oracle* for P is a procedure which, given a point $x \in \mathbb{R}^d$, either certifies that $x \in P$, or returns a valid linear inequality for P that is violated by x. Equivalently, it returns a vector $a \in \mathbb{R}^d$ and a scalar $b \in \mathbb{R}$ such that

$$a^\top y \geq b \qquad \text{for every } y \in P, \qquad \text{but} \qquad a^\top x < b.$$

Theorem 4.17 (Optimization by separation, Grötschel–Lovász–Schrijver)**.** *Let $P \subseteq \mathbb{R}^d$ be a nonempty bounded rational polytope, given by an explicit bounding box and a separation oracle. Given a linear objective $c \in \mathbb{R}^d$ and an accuracy parameter $\varepsilon > 0$, the ellipsoid method computes a point $\widehat{x} \in P$ satisfying*

$$c^\top \widehat{x} \leq \min_{x \in P} c^\top x + \varepsilon$$

using

$$\mathrm{poly}\big(d, B, \log(1/\varepsilon)\big)$$

calls to the separation oracle, where B denotes the relevant bit-complexity of the rational data and the bounding box. Equivalently, weak optimization over a well-described rational polytope is polynomial-time reducible to weak separation; see Grötschel, Lovász and Schrijver [GLS81, GLS88].

We apply Theorem 4.17 to the bounded feasible region

$$P = \left\{ x \in [0,1]^S : \sum_{z \in M(h)} x_z \geq 1 \text{ for every } h \in V^- \right\},$$

over which minimizing $\sum_{z \in S} x_z$ is exactly the LP relaxation above. Note $P \neq \varnothing$, since $x \equiv 1$ is feasible (each $M(h)$ is nonempty); thus P is a nonempty bounded rational polytope with explicit bounding box $[0,1]^S$, as required by the theorem.

The weighted bad-label oracle. For nonnegative weights $w = (w_z)_{z \in S}$, the oracle returns

$$h^\sharp \in \arg\min_{h \in V^-} \sum_{z \in M(h)} w_z.$$

This is a weighted ERM oracle restricted to the bad-label class V^-: it finds the bad-label hypothesis with minimum weighted empirical loss on S.

Proposition 4.18 (Ellipsoid-based oracle algorithm)**.** *Assume an exact weighted bad-label oracle over V^- for rational weights of polynomial bit-complexity, and a known bound $M \geq |\mathcal{M}|$ on the number of distinct mistake sets. Then, for every $\delta \in (0,1)$, a certificate of size $O(\log M) \cdot \mathrm{OPT}$ can be computed with probability at least $1 - \delta$, using $\mathrm{poly}(|S|, B)$ weighted-oracle calls together with $O(\log(1/\delta))$ realizability-oracle verifications, where B is the bit-complexity of the data.*

Proof. Separation oracle. Given $x \in \mathbb{R}^S$, first check the box constraints $0 \leq x_z \leq 1$; a violated one is a separating hyperplane. Otherwise call the weighted bad-label oracle to obtain $h^\sharp \in \arg\min_{h \in V^-} \sum_{z \in M(h)} x_z$. If $\sum_{z \in M(h^\sharp)} x_z < 1$, the constraint indexed by $h^\sharp$ is violated and gives a separating hyperplane; otherwise every covering constraint holds and $x \in P$. Thus the box check together with one oracle call is a separation oracle for P.

Solving the LP. As noted above, P is a nonempty bounded rational polytope with explicit box $[0,1]^S$. Applying Theorem 4.17 to the objective $\sum_{z \in S} x_z$ with accuracy $\varepsilon = 1$ yields, using $\mathrm{poly}(|S|, B)$ oracle calls, a point $\widehat{x} \in P$ with

$$\sum_{z \in S} \widehat{x}_z \;\leq\; \mathrm{LPOpt} + 1 \;\leq\; \mathrm{OPT} + 1 \;\leq\; 2\,\mathrm{OPT},$$

the last step using $\mathrm{OPT} \geq 1$.

Rounding. Since $\widehat{x} \in P$, every edge of the mistake hypergraph $\mathcal{M}$ (at most M of them) satisfies $\sum_{z \in E} \widehat{x}_z \geq 1$. Apply Lemma C.2 with $\mathcal{F} = \mathcal{M}$ and $x = \widehat{x}$: a transversal of $\mathcal{M}$ is exactly a certificate (Proposition 4.3), and the lemma's transversal test is the realizability check "$C \cup \{(x^\star, 1 - y^\star)\}$ unrealizable". The lemma returns, after $O(\log(1/\delta))$ verified trials and with probability at least $1 - \delta$, a certificate of size $O(\log M) \cdot \sum_z \widehat{x}_z = O(\log M) \cdot \mathrm{OPT}$. □

Remark 4.19 (Where the LP barrier sits). The ellipsoid method is used only to avoid enumerating the exponentially many mistake-set constraints. It gives an oracle-efficient way to obtain a fractional LP solution.

The approximation ratio comes later, from rounding. The relevant logarithmic parameter is the number of distinct mistake sets $|\mathcal{M}|$ (only the inclusion-minimal ones need to be hit, so this is conservative). Thus the LP oracle alone does not imply an $O(\log |S|)$-approximation. In the unrestricted case,

$$|\mathcal{M}| \leq 2^{|S|},$$

so the generic logarithmic guarantee is only $O(|S|)$. To obtain an $O(\log |S|)$-type guarantee, one needs an additional structural reason why

$$|\mathcal{M}| \leq \mathrm{poly}(|S|).$$

Bounded VC dimension is one such sufficient condition.

4.6 Bounds

We collect the resulting guarantees and then give the two structural conditions under which the logarithmic scale becomes genuinely sublinear.

4.6.1 Summary of algorithmic guarantees

Write $m = |\mathcal{M}|$ and $n = |S|$. The hypergraph-transversal viewpoint yields two routes to a certificate, sharing the same logarithmic scale.

- *Explicit input.* If $\mathcal{M}$ is given by its incidence matrix, minimum certificate extraction is exactly minimum hitting set. Greedy and LP rounding (Lemma C.2) give an $O(\log m)$-approximation in time $\mathrm{poly}(mn)$; if every edge has size at most r, LP threshold rounding gives an r-approximation (Appendix C).
- *Oracle access.* If $\mathcal{M}$ is not explicit, a weighted bad-label ERM oracle is a separation oracle for the hitting-set LP, and Proposition 4.18 gives an $O(\log M)$-approximation using $\mathrm{poly}(|S|)$ oracle calls, for any known bound $M \geq m$.

Both routes share the same ceiling. In the worst case $m \leq 2^{|S|}$, so the generic ratio is only $O(|S|)$; and since arbitrary finite hypergraphs arise as instances (Proposition 4.9), Feige's set-cover hardness [Fei98] shows that the $(1 - o(1)) \ln m$ scale is essentially optimal under standard complexity assumptions. Any sublinear ratio must therefore come from a structural bound on m, which is the subject of the next two subsections.

4.6.2 Bounded VC dimension

We recall the growth function, the VC dimension, and the Sauer–Shelah lemma, which together bound the number of distinct mistake rows.

Definition 4.20 (Growth function and VC dimension)**.** For a hypothesis class $\mathcal{H} \subseteq \{0,1\}^{\mathcal{X}}$ and $n \in \mathbb{N}$, the *growth function* of $\mathcal{H}$ is

$$\Pi_{\mathcal{H}}(n) \;=\; \max_{x_1, \ldots, x_n \in \mathcal{X}} \big|\{(h(x_1), \ldots, h(x_n)) : h \in \mathcal{H}\}\big|,$$

the largest number of distinct labelings $\mathcal{H}$ realises on n points. A set of n points is *shattered* by $\mathcal{H}$ if $\mathcal{H}$ realises all 2^n labelings on it, and the *VC dimension* $\mathrm{VCdim}(\mathcal{H})$ is the largest n for which some n-point set is shattered (∞ if arbitrarily large sets are shattered).

Lemma 4.21 (Sauer–Shelah [Sau72, She72])**.** *If* $\mathrm{VCdim}(\mathcal{H}) = d < \infty$*, then for every* $n \geq 1$*,*

$$\Pi_{\mathcal{H}}(n) \;\leq\; \sum_{i=0}^{d} \binom{n}{i}.$$

In particular, for $n \geq d \geq 1$*,*

$$\Pi_{\mathcal{H}}(n) \;\leq\; \left(\frac{e\,n}{d}\right)^{d} \;=\; O(n^{\,d}),$$

so the growth function is polynomial in n *of degree* d*.*

Proposition 4.22 (Bounded VC dimension yields a logarithmic approximation)**.** *If* $\mathrm{VCdim}(\mathcal{H}) = d < \infty$*, then the mistake hypergraph satisfies*

$$|\mathcal{M}| \;\leq\; \sum_{i=0}^{d} \binom{|S|}{i} \;\leq\; \left(\frac{e|S|}{d}\right)^{d} \;=\; O(|S|^{\,d}).$$

Consequently, the oracle algorithm of Proposition 4.18, applied with the bound $M = O(|S|^d)$*, computes an* $O(d \log |S|)$*-approximation to the minimum certificate problem using* $\mathrm{poly}(|S|)$ *oracle calls; for constant* d *this is an* $O(\log |S|)$*-approximation.*

Proof. Fixing a target labeling on S, the family of mistake sets

$$\{\{z \in S : h(z_x) \neq z_y\} : h \in \mathcal{H}\}$$

is the trace on the $|S|$ sample points of the class obtained from $\mathcal{H}$ by relabeling against the target; this relabeling is a bijection on labelings and hence preserves the growth function. Restricting further to the bad-label subclass V^- cannot increase the number of traces. Hence, by Lemma 4.21,

$$|\mathcal{M}| \;\leq\; \left|\{M(h) : h \in V^-\}\right| \;\leq\; \sum_{i=0}^{d} \binom{|S|}{i} \;\leq\; \left(\frac{e|S|}{d}\right)^{d} \;=\; O(|S|^{\,d}).$$

Plugging this bound on $|\mathcal{M}|$ into Proposition 4.18 (with $M = O(|S|^d)$, so that $\log M = O(d \log |S|)$) gives the stated $O(d \log |S|)$-approximation; for constant d the ratio is $O(\log |S|)$. □

The important point is that this bound comes from controlling the number of distinct rows. Without such a growth bound, $|\mathcal{M}|$ may be exponential in $|S|$, and the generic logarithmic guarantee becomes only $O(|S|)$.

Remark 4.23 (Open Question: Is VC dimension a necessary condition?)*.* Using the Sauer–Shelah lemma (Lemma 4.21), we showed that bounded VC dimension is one sufficient way to obtain an upper bound on the number of distinct mistake rows, and hence a logarithmic-in-$|S|$ approximation. Whether some comparable growth condition is necessary for this LP-based route remains to be established.

4.6.3 Bounded hollow star number

The hollow star number s_0 controls the largest possible size of a minimal-inclusive certificate [BHPS25], which yields both an approximation guarantee and, for small s_0, an exact algorithm.

Proposition 4.24 (Hollow star number bounds the optimum)**.** OPT $\leq s_0(\mathcal{H})$. *Consequently, the deletion procedure of Proposition 4.1 is an s_0-approximation algorithm for the minimum certificate problem in the realizable setting.*

Proof. Every minimum certificate is in particular inclusion-minimal, and the worst-case size of a minimal-inclusive certificate is at most s_0 [BHPS25]; hence OPT $\leq s_0$. The procedure of Proposition 4.1 returns a minimal-inclusive certificate, of size at most $s_0 \leq s_0 \cdot \mathrm{OPT}$, using OPT ≥ 1 in the nontrivial case $V^- \neq \varnothing$. □

Proposition 4.25 (Exact extraction for small s_0)**.** *The minimum certificate problem can be solved exactly using at most*

$$\sum_{i=0}^{s_0} \binom{|S|}{i} \;\leq\; |S|^{s_0}$$

certificate-verification calls. In particular, for fixed s_0 it is exactly solvable in time polynomial in $|S|$.

Proof. By Proposition 4.24, OPT $\leq s_0$, so it suffices to enumerate all subsets $C \subseteq S$ of size at most s_0, test each for being a certificate (the realizability check of Proposition 4.1), and output the smallest certificate found. The number of subsets tested is $\sum_{i=0}^{s_0} \binom{|S|}{i} \leq |S|^{s_0}$. □

More generally, exact enumeration is useful whenever $s_0 \ll |S|$, although the exponent still depends on s_0.

5 Conclusion

We studied two structural questions about robust certificates, both of which reduce to transversal-type problems on hypergraphs derived from a hypothesis class. *Combinatorially*, recasting b-robust realizability as a b-tolerant monotone property and applying the tolerance Helly framework of Montejano and Oliveros, together with Tuza's extremal estimate, gives an upper bound on the b-robust hollow star number s_b in terms of s_0 and the corruption budget b (Section 3). *Algorithmically*, minimum certificate extraction is exactly minimum hypergraph transversal, in both directions, which transfers the structural and approximation theory of hitting set to our setting — including an $O(\log |\mathcal{M}|)$-approximation through a weighted-ERM separation oracle and its $O(d \log |S|)$ specialisation under bounded VC dimension (Section 4). The precise statements and their consequences are collected in the introduction; here we only stress the common thread: a single triplet $(\mathcal{H}, S, x^\star)$ admits different hypergraph encodings, each natural to the question at hand, and in both encodings the certificate question becomes a transversal question. The remaining gaps — chiefly the algorithmic side of the b-robust setting and an active learning *certifier* variant are discussed in 6.

6 Further directions

We close with directions that seem most directly connected to the results above. The first is a formal extension of the algorithmic equivalence of Section 4; the second concerns the sampling and active-query process used to obtain the initial certificate.

6.1 Completing the algorithmic picture

Section 4 is developed in the realizable setting $b = 0$, whereas the combinatorial bound of Section 3 already applies to general b. A natural next step is therefore to formulate and analyse the extraction problem for $b > 0$.

The expected analogue is a transversal problem with multiplicity. In the realizable case, a certificate must hit every bad-label mistake set at least once. In the b-robust case, a bad-label hypothesis should be ruled out even after up to b corruptions, suggesting that each relevant mistake set must be hit at least $b+1$ times. Making this statement precise requires care: one must specify the robust bad-label version space, decide whether to prune to a clutter before or after imposing multiplicities, and check that the resulting optimization problem exactly matches the certificate definition.

Two further algorithmic questions remain open.

1. The LP-based algorithm has an $\Omega(\log|\mathcal{M}|)$ barrier for this relaxation, coming from the standard set-cover integrality gap. This does not by itself prove that every algorithm with access to the same learning oracles must incur such a cost.

2. Bounded VC dimension gives one sufficient route to an upper bound on $|\mathcal{M}|$, and therefore to an $O(\log|S|)$-type approximation. It remains open whether a comparable growth condition is necessary for this particular LP/separation-oracle approach, or whether weaker structure of the bad-label traces suffices.

6.2 Active certification: improved query complexity

A second question, complementary to certificate extraction, concerns how the initial certifying sample is obtained. In the passive sampling procedure of [BHPS25], described in Section 4.4, the certifier draws labeled examples i.i.d. from a distribution and stops once the accumulated sample certifies the fixed test point $x^\star$.

The active variant is potentially more efficient: the certifier first observes an unlabeled point x drawn from the marginal distribution and then decides whether to query its label, subject to a requirement that it queries often enough. The central question is whether active querying can reduce the number of labels needed to certify $x^\star$, or produce initial certificates whose induced mistake hypergraphs are easier to sparsify.

One concrete direction is to compare this process with the standard CAL (Cohn–Atlas–Ladner) algorithm [CAL94]. Under distributional assumptions such as log-concavity, one may hope that querying only inside an appropriate disagreement region reduces the label complexity of certification. A second, more structural, direction is to ask whether active querying can bias the observed sample toward instances whose mistake hypergraphs admit better approximation guarantees than those available in the worst-case static model.

Acknowledgements

The elementary critical-set bound of Appendix A is due to Amartya Sanyal, whom I thank for this argument and for several helpful discussions.

A An elementary bound on s_b via critical sets

This proof is due to Amartya Sanyal. The bound of Section 3 routes through the tolerance framework of Montejano and Oliveros. For completeness we record this alternative, self-contained argument that avoids the Helly-type machinery entirely: it uses only the definition of the hollow star number s_0. As we show in §A.3, however, the bound it yields is asymptotically looser in the regime $b \gg s_0$.

Theorem A.1. *For every hypothesis class $\mathcal{H}$ and every integer $b \geq 0$,*

$$s_b(\mathcal{H}) \;\leq\; 1 + \sum_{t=1}^{b+1} s_0(\mathcal{H})^{\,t}.$$

The proof has two parts. We first introduce the *critical sets* of an unrealizable sequence and bound how many indices can participate in a t-element critical set; we then show that a b-robust hollow star set injects its light points into the $(b{+}1)$-critical indices of its underlying unweighted sequence.

A.1 Critical sets

Let $A = ((x_1, y_1), \ldots, (x_n, y_n))$ be an unrealizable labeled sequence. For $I \subseteq [n]$, write $A \setminus I$ for the subsequence obtained by deleting the entries whose indices lie in I, and $A[I]$ for the subsequence keeping exactly those entries. For an integer $t \geq 1$, a set $I \subseteq [n]$ is *t-critical* (for A) if

(i) $|I| = t$,
(ii) $A \setminus I$ is realizable, and
(iii) $A \setminus J$ is unrealizable for every proper subset $J \subsetneq I$.

Let $\mathrm{Crit}_t(A) = \{\, i \in [n] : i \text{ lies in some } t\text{-critical set of } A \,\}$ and $c_t(A) = |\,\mathrm{Crit}_t(A)|$, and set

$$C_t(\mathcal{H}) \;=\; \sup\{\, c_t(A) : A \text{ is unrealizable by } \mathcal{H} \,\}.$$

Throughout this section we abbreviate $s := s_0(\mathcal{H})$.

Lemma A.2. *$C_1(\mathcal{H}) \leq s$.*

Proof. Fix an unrealizable sequence A and choose an inclusion-minimal set of indices $M \subseteq [n]$ for which $A[M]$ is unrealizable. For each $r \in M$, minimality makes $A[M \setminus \{r\}]$ realizable; any h realizing it must satisfy $h(x_r) \neq y_r$, since otherwise h would realize all of $A[M]$. Hence every single-index deletion from $A[M]$ is realizable, i.e. $A[M]$ is a hollow star set (Definition 2.3), so $|M| \leq s$.

If $i \notin M$, then $A \setminus \{i\}$ still contains the unrealizable subsequence $A[M]$, so $A \setminus \{i\}$ is unrealizable and i is not 1-critical. Thus $\mathrm{Crit}_1(A) \subseteq M$ and $c_1(A) \leq |M| \leq s$. Taking the supremum over A gives $C_1(\mathcal{H}) \leq s$. □

Lemma A.3. *For every $t \geq 2$, $\quad C_t(\mathcal{H}) \leq s + s\, C_{t-1}(\mathcal{H})$.*

Proof. Fix an unrealizable A and an inclusion-minimal $M \subseteq [n]$ with $A[M]$ unrealizable; as in Lemma A.2, $A[M]$ is a hollow star set, so $|M| \leq s$. Take any index $i \in \mathrm{Crit}_t(A) \setminus M$, lying in some t-critical set I. Since $A \setminus I$ is realizable, I must meet M: otherwise the unrealizable $A[M]$ would survive in $A \setminus I$, a contradiction. Fix $r \in I \cap M$. As $\{r\} \subsetneq I$ and I is t-critical, $A \setminus \{r\}$ is unrealizable, and we claim $I \setminus \{r\}$ is $(t{-}1)$-critical for $A \setminus \{r\}$. Indeed,

$$(A \setminus \{r\}) \setminus (I \setminus \{r\}) \;=\; A \setminus I \quad \text{is realizable,}$$

while for any $J \subsetneq I \setminus \{r\}$ we have $(A \setminus \{r\}) \setminus J = A \setminus (J \cup \{r\})$ with $J \cup \{r\} \subsetneq I$, which is unrealizable by the t-criticality of I. Hence $i \in \mathrm{Crit}_{t-1}(A \setminus \{r\})$, and therefore

$$\mathrm{Crit}_t(A) \setminus M \;\subseteq\; \bigcup_{r \in M} \mathrm{Crit}_{t-1}(A \setminus \{r\}).$$

Consequently $c_t(A) \leq |M| + \sum_{r \in M} c_{t-1}(A \setminus \{r\}) \leq s + s\, C_{t-1}(\mathcal{H})$. Taking the supremum over all unrealizable A proves the claim. □

Corollary A.4. *For every $t \geq 1$, $\quad C_t(\mathcal{H}) \leq \displaystyle\sum_{j=1}^{t} s^j$.*

Proof. Induction on t. The base case $t = 1$ is Lemma A.2. For $t \geq 2$, Lemma A.3 and the inductive hypothesis give

$$C_t(\mathcal{H}) \;\leq\; s + s \sum_{j=1}^{t-1} s^j \;=\; \sum_{j=1}^{t} s^j. \qquad \square$$

A.2 From robust hollow stars to critical sets

Let T be a b-robust hollow star set for $\mathcal{H}$ of size m (Definition 2.5), with star index $i^\star$ of weight $b{+}1$ and the remaining $m-1$ *light* points of weight 1. Let U be the underlying *unweighted* sequence on the same m points.

First, U is unrealizable: a hypothesis realizing U would incur weighted error 0 on T, contradicting condition (ii) of Definition 2.5.

Now fix a light index i. By condition (iii), $T \setminus \{i\}$ is b-robustly realizable, so some h_i has weighted error at most b on it. Since the star point has weight $b{+}1$, h_i must classify it correctly, hence misclassifies at most b of the remaining light points. In fact it misclassifies *exactly* b of them: if it erred on only $r < b$ light points $j \neq i$, then on the full T its weighted error would be at most $r + 1 \leq b$ (the extra 1 accounting for the omitted index i), contradicting condition (ii). Let J_i be these b misclassified light indices and put $K_i = \{i\} \cup J_i$, so $|K_i| = b + 1$ and h_i realizes $U \setminus K_i$.

We claim K_i is $(b{+}1)$-critical for U. We already have $U \setminus K_i$ realizable. For any proper subset $R \subsetneq K_i$ the star point survives in $U \setminus R$, so any hypothesis realizing $U \setminus R$ classifies it correctly and disagrees with U only on the light indices in R, incurring weighted error at most $|R| \leq b$ on T — again contradicting condition (ii). Hence $U \setminus R$ is unrealizable for every $R \subsetneq K_i$, and K_i is indeed $(b{+}1)$-critical.

Thus every light index lies in $\mathrm{Crit}_{b+1}(U)$, so $m - 1 \leq c_{b+1}(U) \leq C_{b+1}(\mathcal{H})$, i.e. the size of T satisfies $m \leq 1 + C_{b+1}(\mathcal{H})$. Taking the supremum over all b-robust hollow star sets and applying Corollary A.4,

$$s_b(\mathcal{H}) \;\leq\; 1 + C_{b+1}(\mathcal{H}) \;\leq\; 1 + \sum_{t=1}^{b+1} s_0(\mathcal{H})^{\,t},$$

which proves Theorem A.1. □

A.3 Comparison of the two bounds

Theorem 3.6 and Theorem A.1 give two upper bounds on s_b in terms of s_0:

$$\text{(Tolerance Helly / Tuza)} \qquad s_b \;\leq\; \eta(s_0,\, b{+}1) \;<\; \binom{s_0 + b}{s_0 - 1} + \binom{s_0 + b - 1}{s_0 - 1},$$

$$\text{(critical sets)} \qquad s_b \;\leq\; 1 + \sum_{t=1}^{b+1} s_0^{\,t}.$$

The binomial $\binom{s_0+b}{s_0-1}$ is a polynomial of degree $s_0 - 1$ in b and of degree $b + 1$ in s_0, whereas the geometric sum satisfies $\sum_{t=1}^{b+1} s_0^{\,t} = \Theta(s_0^{\,b+1})$. Which bound is sharper depends on which parameter is held fixed.

Fixed b, $s_0 \to \infty$. Both bounds are $\Theta(s_0^{\,b+1})$, differing only in the leading constant.

Fixed s_0, $b \to \infty$. The Tolerance-Helly bound is $O(b^{\,s_0-1})$ — polynomial in b — whereas the critical-set bound is $\Theta(s_0^{\,b+1})$ — exponential in b. The elementary bound is therefore far looser once $b \gg s_0$.

The case $b = 1$. Both bounds are quadratic in s_0, but with different leading constants:

$$\text{Tuza:} \quad s_1 < \binom{s_0 + 1}{2} + \binom{s_0}{1} = \tfrac{1}{2}s_0^2 + \tfrac{3}{2}s_0, \qquad \text{critical sets:} \quad s_1 \leq 1 + s_0 + s_0^2.$$

The leading coefficients are $\frac{1}{2}$ and 1, so even at $b = 1$ the elementary bound is looser by a factor of 2 in the dominant term. It remains open to determine the sharp leading constant, and more

generally to understand when the Tolerance-Helly bound is tight for hypothesis classes arising in learning theory.

B The hollow star number as a finite dual Helly number

We make precise the claim, used informally in Sections 1 and 2, that the hollow star number is the dual Helly number with its defining quantifier restricted to *finite* collections. Recall (Definition 2.2) that the dual Helly number $k_w(\mathcal{H})$ is the smallest k such that *every* (possibly infinite) $S \subseteq \mathcal{X} \times \mathcal{Y}$ with $V(S) = \varnothing$ admits a witness $W \subseteq S$ with $|W| \leq k$ and $V(W) = \varnothing$. Restricting the quantifier to finite S defines the *finite-restricted dual Helly number* $k_w^{\text{fin}}(\mathcal{H})$: the smallest k such that every *finite* $S \subseteq \mathcal{X} \times \mathcal{Y}$ with $V(S) = \varnothing$ has a witness $W \subseteq S$ with $|W| \leq k$ and $V(W) = \varnothing$ (and $k_w^{\text{fin}}(\mathcal{H}) = \infty$ if no finite k works). Throughout we use that $V(W) = \bigcap_{(x,y)\in W} V(x, y)$, so $V(W) = \varnothing$ is exactly unrealizability of W (Section 2).

Proposition B.1. *For every hypothesis class $\mathcal{H}$, $k_w^{\text{fin}}(\mathcal{H}) = s_0(\mathcal{H})$.*

Proof. $s_0(\mathcal{H}) \leq k_w^{\text{fin}}(\mathcal{H})$. Let S be a hollow star set (Definition 2.3) of size m. It is finite and unrealizable, so $V(S) = \varnothing$, while every *proper* subset of S is realizable. Hence the *only* subset $W \subseteq S$ with $V(W) = \varnothing$ is $W = S$ itself, of size m. For the defining condition of k_w^{fin} to hold for this S, the bound k must satisfy $k \geq m$. As this holds for every hollow star set, $k_w^{\text{fin}}(\mathcal{H}) \geq s_0(\mathcal{H})$.

$k_w^{\text{fin}}(\mathcal{H}) \leq s_0(\mathcal{H})$. Let S be any finite collection with $V(S) = \varnothing$, i.e. unrealizable. Since S is finite, it has an inclusion-minimal unrealizable subset $W \subseteq S$. By minimality every proper subset of W is realizable, so W is a hollow star set and $|W| \leq s_0(\mathcal{H})$; moreover $V(W) = \varnothing$. Thus $k = s_0(\mathcal{H})$ witnesses the defining condition for every finite unrealizable S, giving $k_w^{\text{fin}}(\mathcal{H}) \leq s_0(\mathcal{H})$. □

The restriction to finite S is essential: the unrestricted k_w may be strictly larger — indeed infinite while s_0 is finite, as for singletons over an infinite domain (Remark 2.4). The same identity also follows from the monotone-property viewpoint of Section 3.2, where the Helly number of realizability is shown to be $s_0(\mathcal{H})$.

C Hitting set, set cover, and randomized rounding

This appendix collects the classical explicit-input facts about minimum hitting set used in Section 4, together with the one rounding statement (Lemma C.2) that the oracle algorithm of §4.5 reuses. Recall (Proposition 4.8) that minimum certificate extraction is exactly minimum transversal of the mistake hypergraph; here we treat the abstract problem on a general hypergraph and assume the input is given explicitly.

C.1 Setup and duality

Let $G = (V, E)$ be a hypergraph. The *hypergraph vertex-cover* (equivalently, *hitting-set* or *transversal*) problem is

$$\min\{|C| : C \subseteq V,\ C \cap e \neq \varnothing \text{ for every } e \in E\}.$$

We write $n = |V|$, $m = |E|$, vertex degree $d(v) = |\{e \in E : v \in e\}|$, maximum degree $\Delta(G) = \max_v d(v)$, and rank $r(G) = \max_{e\in E} |e|$. Throughout we assume the input is the full incidence matrix $A_{e,v} = \mathbf{1}[v \in e]$, of size $\Theta(mn)$.

The problem is equivalent to set cover by duality: take the universe $U = E$ and one set $A_v = \{e \in E : v \in e\}$ per vertex $v \in V$; then a set of vertices $C \subseteq V$ hits every hyperedge iff $\{A_v : v \in C\}$ covers U. Equivalently, the dual set-cover instance is encoded by the transpose incidence matrix, so hypergraph transversal and set cover are the same optimization problem.

C.2 Greedy

Greedy repeatedly chooses a vertex hitting the largest number of currently unhit edges (equivalently, a set A_v covering the largest number of uncovered universe elements). With incidence-matrix access, a naive implementation runs in time $\mathrm{poly}(mn)$.

Proposition C.1 (Greedy guarantee)**.** *Greedy returns a hitting set of size at most* $H_{\Delta(G)} \cdot \mathrm{OPT} \le (1+\ln \Delta(G))\,\mathrm{OPT}$*, where* $\Delta(G)$ *is the maximum vertex degree and* OPT *is the minimum hitting-set size. In particular it is an* $O(\log m)$*-approximation.*

Proof. Let OPT be the size of an optimum cover. Suppose that at some step there are q uncovered elements of U. An optimum cover uses OPT sets to cover all elements, so it contains some set covering at least q/OPT of the currently uncovered elements; greedy chooses a set covering at least as many. Hence one greedy step drops the number of uncovered elements from q to at most $q(1-\frac{1}{\mathrm{OPT}})$. After t steps the number of uncovered elements is at most

$$m\left(1-\frac{1}{\mathrm{OPT}}\right)^{t} \le m\,e^{-t/\mathrm{OPT}},$$

which is below 1 once $t \ge \mathrm{OPT}\ln m$. Thus greedy terminates after $O(\mathrm{OPT}\log m)$ steps, an $O(\log m)$-approximation.

The sharper harmonic bound follows from a charging argument: when greedy picks a set covering k new elements, charge each a cost $1/k$; the total cost equals the number of sets chosen. Since within any single optimum set the elements are covered in some order and the j-th-from-last covered element is charged at most $1/j$, each optimum set absorbs total charge at most $H_{\Delta(G)} = \sum_{j=1}^{\Delta(G)} 1/j \le 1+\ln\Delta(G)$. Summing over the OPT optimum sets bounds the number of sets greedy chooses by $H_{\Delta(G)} \cdot \mathrm{OPT}$. □

C.3 LP relaxation and rounding

The integer program for hypergraph transversal is $\min \sum_{v\in V} x_v$ subject to $\sum_{v\in e} x_v \ge 1$ for every $e \in E$ and $x_v \in \{0,1\}$; its standard relaxation replaces $x_v \in \{0,1\}$ by $x_v \ge 0$. Two basic guarantees follow. If the hypergraph has bounded rank $r(G)$, the threshold rule $C = \{v : x_v \ge 1/r(G)\}$ gives an $r(G)$-approximation: every edge e has $\sum_{v\in e} x_v \ge 1$ and $|e| \le r(G)$, so some $v \in e$ has $x_v \ge 1/r(G)$. The general route is randomized rounding, which gives an $O(\log m)$-approximation. We state it in the form reused in the main text, phrased so that it applies to *any* fractional point feasible for the covering constraints — both to an exact LP optimum and to the near-optimal fractional point produced by the ellipsoid method (§4.5).

Lemma C.2 (Randomized rounding for hitting set)**.** *Let* $\mathcal{F}$ *be a hypergraph on a finite ground set* V *with* $|\mathcal{F}| \le M$ *edges, and let* $x \in [0,1]^V$ *be a point with* $\sum_{v\in e} x_v \ge 1$ *for every* $e \in \mathcal{F}$*. Put* $\alpha = \ln(2M)$ *and form a random set* $C \subseteq V$ *by including each* $v \in V$ *independently with probability* $\min\{1, \alpha x_v\}$*. Then*
(i) $\Pr[\,C$ *is not a transversal of* $\mathcal{F}\,] \le \frac{1}{2}$*, and*
(ii) $\mathbb{E}[\,|C|\,] \le \alpha \sum_{v\in V} x_v$*.*
Consequently, running $O(\log(1/\delta))$ *independent trials and keeping the smallest set that is a transversal yields, with probability at least* $1-\delta$*, a transversal of size* $O(\log M)\cdot\sum_{v\in V} x_v$*.*

The $O(\log M)$ loss is tight for the standard set-cover LP relaxation in the following LP-based sense. Lovász [Lov75] showed that the ratio between the optimum integral cover and the optimum fractional cover can be as large as the relevant harmonic number; in particular, for a universe of size M, the integrality gap of the standard set-cover LP is $\Theta(\log M)$. Since our hitting-set formulation is exactly the dual set-cover formulation, with universe equal to the set of mistake hyperedges, this lower bound applies here as well. Consequently, no rounding theorem whose guarantee is measured only against the value of this standard LP relaxation can guarantee an $o(\log M)$ factor in the unrestricted worst case.

Proof. For an edge $e \in \mathcal{F}$, using $1 - p \le e^{-p}$,

$$\Pr[C \cap e = \varnothing] = \prod_{v \in e} (1 - \min\{1, \alpha x_v\}) \le \prod_{v \in e} e^{-\alpha x_v} = e^{-\alpha \sum_{v \in e} x_v} \le e^{-\alpha} = \frac{1}{2M},$$

and a union bound over the at most M edges gives (i). Claim (ii) is immediate: $\mathbb{E}[|C|] = \sum_v \min\{1, \alpha x_v\} \le \alpha \sum_v x_v$. For the last statement, Markov's inequality applied to (ii) gives $\Pr[\,|C| > 4\alpha \sum_v x_v\,] \le \frac{1}{4}$, so by (i) a single trial is, with probability at least $\frac{1}{4}$, simultaneously a transversal and of size at most $4\alpha \sum_v x_v$. Repeating independently $O(\log(1/\delta))$ times and returning the smallest transversal found succeeds with probability at least $1 - \delta$; since $\alpha = O(\log M)$, the output has size $O(\log M) \cdot \sum_v x_v$. □

Applying Lemma C.2 to an optimal LP solution ($\mathcal{F} = E$, $M = m$, $\sum_v x_v^\star = L \le \mathrm{OPT}$) gives the explicit-input $O(\log m)$-approximation, matching the greedy scale.

C.4 References and tightness

The logarithmic guarantee for set cover is classical; see Johnson [Joh74], Lovász [Lov75], and Chvátal [Chv79]. Feige [Fei98] proved that the $(1 - o(1)) \ln m$ scale is essentially optimal for polynomial-time approximation under standard complexity assumptions, where m is the universe size — in our hitting-set formulation, the number of distinct mistake sets.

Finally we record the elementary brute-force baseline quoted in §4.5, which shows that beating the trivial $|S|$-approximation by any constant factor is easy and so not the right target.

Proposition C.3 (Brute-force baseline)**.** *For every fixed constant $\rho \in (0, 1]$, there is a polynomial-time algorithm, using only certificate verification, that achieves a $\rho|S|$-approximation for the minimum certificate problem.*

Proof. Let $k = \lceil 1/\rho \rceil - 1$. Enumerate all subsets $C \subseteq S$ of size at most k and test each for being a certificate; if one is found, output the smallest such certificate, and otherwise output all of S. If no certificate of size at most k exists then $\mathrm{OPT} \ge k + 1 \ge 1/\rho$, so outputting S gives $|S|/\mathrm{OPT} \le \rho|S|$. The enumeration uses $\sum_{i=0}^{k} \binom{|S|}{i} = O(|S|^k)$ verification calls, which is polynomial in $|S|$ for fixed ρ. □

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Declaration of using generative AI tools (for students)

*At the University of Copenhagen, we do our work with **responsibility** and **respect for society, cultural heritage, the environment and people** around us.*
***Integrity, honesty** and **transparency** are prerequisites for academic work. We expect exam performances to reflect the **student's own learning and independent work**.*
*Academic work is always based on other people's insights, knowledge and contributions, but always with thorough **recognition, respect** and **crediting** of this work. This also applies when using generative artificial intelligence.*

Guidelines

Use of generative AI in exams

According to the University's rules for using tools based on generative AI (GAI), you must be transparent about your use of the technology, for example in your methodology section and/or by filling in and submitting the declaration template below as an appendix to your exam papers. When you fill in the declaration, it is important that the reader gets a clear picture of whether and how generative AI has contributed to the final product.

If the template has been chosen as the preferred means of declaration in your course, you must also use it when you have *not* used GAI tools as an aid. In this case, the only thing you need to do is tick the box stating that you have not used generative AI, leaving the rest blank.
By declaring your use of GAI tools, you avoid issues in relation to the rules on exam cheating.

On courses where the use of GAI has been an integrated part of the content, reflection and critical assessment of using GAI may also be included in your exam paper's methodology section. If you are unsure of whether this is the case on your course, you can ask your lecturer or supervisor.

If GAI is the object of your exam paper, its role will appear from your research questions, methodology description, analysis and conclusion. If you also use GAI as an aid in the process, you need to declare it separately.

Points of attention:

- If GAI is a permitted aid on your course, you can use it for dialogue and sparring when writing your exam paper, but you are **not** allowed to let GAI write your paper, even if all aids are permitted.
- If you include GAI material as a source (directly or edited) in your exam paper, the same requirements apply for using quotation marks and source referencing as with all other sources. Otherwise, it will be a case of plagiarism.
- Never feed an AI tool with data that are personally identifiable, protected by copyright or confidential.
- Always remember to check the current rules and guidelines for using GAI at UCPH.
- Read the course description carefully. It's important that you know which aids are permitted on your course. There may be additional documentation requirements, for example that you must describe your key prompts and any source material (what context you have given, what you have fed into the tool, what you have asked the tool to do), describe the output (what responses did the tool give you?), describe the process, for example history and iterations (if you have written back and forth with the tool several times to get a useful output).
- If in doubt, talk to your lecturer or supervisor.

Declaration of using generative AI tools (for students)

☒ **I/we have used generative AI as an aid/tool** *(please tick)*

☐ **I/we have <u>NOT</u> used generative AI as an aid/tool** *(please tick)*

If generative AI is permitted in the exam, but you haven't used it in your exam paper, you just need to tick the box stating that you have not used GAI. You don't have to fill in the rest.

List which GAI tools you have used and include the link to the platform (if possible):

ChatGPT

Describe how generative AI has been used in the exam paper:

1) *Purpose (what did you use the tool for?)*
2) *Work phase (when in the process did you use GAI?)*
3) *What did you do with the output? (including any editing of or continued work on the output)*

I used ChatGPT mainly for surveying literature, a digital teacher for topics I was not familiar with, discussing ideas, and refining my suggestions. The second use was in formatting word files into latex formats, as well as making sure I did not mix up notation.
I.e. GAI was used continuously both during the research phase, as well as a formatting tool. The output of it was either heavily refined or completely replaced.

I included a reference to ChatGPT in my Thesis, and cited it clearly as a contributor in one part of my work where it also partially suggested the mathematical idea.

Please note: Content generated by GAI that is used as a source in the paper requires correct use of quotation marks and source referencing. Read the guidelines from Copenhagen University Library at KUnet.